



**Northern Beaches Secondary College
Manly Selective Campus**

**2010
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators and templates may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1- 8
- All questions are of equal value

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Marks

Question 1 (Answer in a separate booklet)

(15)

(a) Find $\int \frac{1}{\sqrt{4x^2 - 25}} dx$ (2)

(b) Find $\int \tan^4 x \sec^2 x dx$ (2)

(c) Find $\int_0^{\frac{\pi}{2}} x \sin x dx$ (3)

(d) Use the substitution $u = x-2$ to evaluate $\int_{\frac{3}{2}}^{\frac{5}{2}} \frac{1}{\sqrt{(x-1)(3-x)}} dx$ (3)

(e) (i) Write $\frac{3x+2}{x^2+5x+6}$ as a sum of partial fractions. (2)

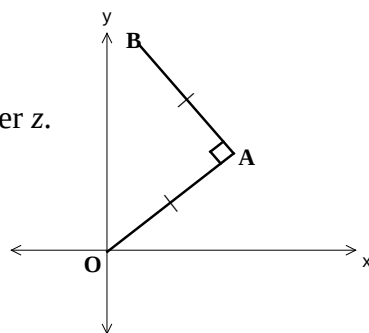
(ii) Hence evaluate $\int_0^2 \frac{3x+2}{x^2+5x+6} dx$ (3)

Question 2 (Answer in a separate booklet)

(15)

(a) Find the value of z^{10} in the form $x + iy$ when $z = \sqrt{2} - \sqrt{2}i$ (2)

(b) The complex vector z is represented in the accompanying diagram by the point A.
 The triangle OAB is a right angled isosceles triangle.



(i) Express the point B in terms of the complex number z . (1)

(ii) Let M be the midpoint of AB.
 What complex number corresponds to M? (1)

(iii) Give an example of the vector C which
 would make a trapezium? (1)

(c) (i) Write down the value of i^6 . (1)

(ii) Hence or otherwise plot and label the sixth roots of -1 about the unit circle. (2)

(iii) Find the roots of $x^4 - x^2 + 1 = 0$ in modulus-argument form. (2)

(d) Given the locus of z is $|z - 2 - 2i| = 1$

(i) Sketch the locus of z on the argand diagram. (1)

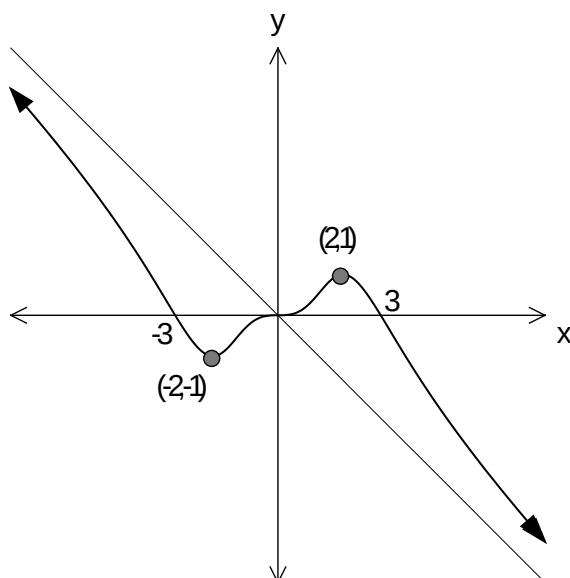
(ii) Find the maximum value of $\arg z$. (2)

(iii) Find the maximum value of $\text{mod } z$. (2)

Question 3 (Answer in a separate booklet)

(15)

(a) The graph of $y = f(x)$ is shown. The line $y = -x$ is an oblique asymptote to the curve.



Use separate half page graphs, to sketch

(i) $f(-x)$ (1)

(ii) $f(|x|)$ (1)

(iii) $\frac{x}{f(x)}$ (3)

(b) Given that $1 + i$ is a zero of $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$, factorise $P(x)$ fully over the field of the complex numbers. (2)

(c) The equation $x^3 + x^2 - 2x - 3 = 0$ has roots α , β and γ . Find the equation with roots:

(i) α^2 , β^2 and γ^2 (1)

(ii) $\alpha^3\beta\gamma$, $\alpha\beta^3\gamma$ and $\alpha\beta\gamma^3$ (2)

(d) For the hyperbola $\frac{x^2}{16} - \frac{y^2}{25} = 1$,

(i) find the eccentricity. (1)

(ii) find the coordinates of the foci. (1)

(iii) find the equations of the directrices. (1)

(iv) find the equations of the asymptotes. (1)

(v) if P is on the hyperbola and S and S' are it's foci, then given PS=2, find PS'. (1)

Question 4 (Answer in a separate booklet)

(15)

- (a) Using the method of cylindrical shells, find the volume generated by revolving the region bounded by $y = \log_e x$, the x -axis and the interval $1 \leq x \leq e$ about the y axis.

(4)

- (b) The ellipse $\frac{(x-4)^2}{9} + \frac{y^2}{4} = 1$ is rotated about the y axis to form a donut shape.

- (i) By taking slices perpendicular to the axis of rotation, show that the

volume of a slice is $8\pi \sqrt{36 - 9y^2} \delta y$ (2)

- (ii) Find the volume of the solid (2)

- (c) Calculate the modulus and argument of the sum of the roots of the equation (2)

$$(3 + 4i)z^2 + (2 - i)z + (8 - 2i) = 0$$

- (d) PQ is a chord of a rectangular hyperbola $xy = c^2$

- (i) Show that PQ has equation $x + pqy = c(p+q)$ where P and Q have parameters p and q respectively. (2)

- (ii) If PQ has a constant length k^2 , show that

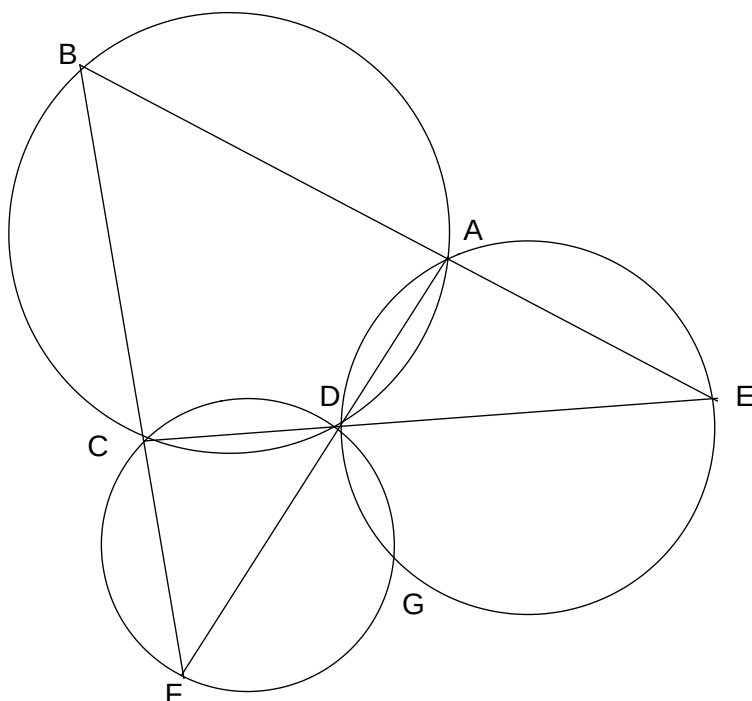
$$c^2[(p+q)^2 - 4pq](p^2q^2 + 1) = k^4p^2q^2$$

and find the locus of R, the midpoint of PQ, in Cartesian form. (3)

Question 5 (Answer in a separate booklet)

(15)

- (a) ABCD is a cyclic quadrilateral. BA and CD are both produced and intersect at E. BC and AD produced intersect at F. The circles EAD and FCD intersect at G as well as at D. Prove the points E, G and F are collinear. (4)



- (b) Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ for integers n , $n \geq 0$

(i) Show that $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$ for $n \geq 2$. (2)

(ii) Deduce that $I_n = \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. (2)

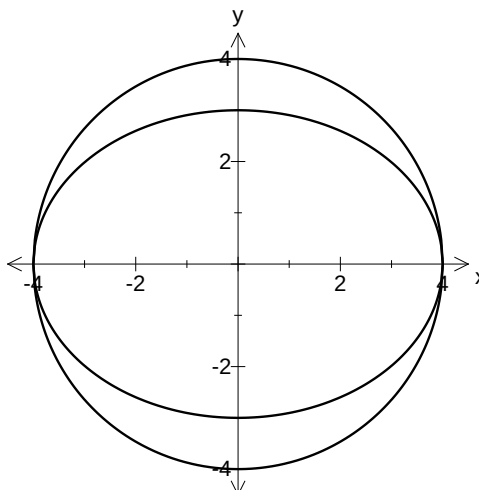
(iii) Evaluate I_4 (2)

- (c) (i) If α is a multiple root of the polynomial equation $P(x) = 0$, prove that $P'(\alpha) = 0$ (2)
- (ii) Find all roots of the equation $18x^3 + 3x^2 - 28x + 12 = 0$ if two of the roots are equal. (3)

Question 6 (Answer in a separate booklet)

(15)

- (a) The diagram shows an ellipse with equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the larger of its auxiliary circles. The coordinates of a point P on the ellipse are $(4\cos \theta, 3\sin \theta)$ where $\theta \neq 0$ or π .



A straight line l parallel to the y axis intersects the x axis at N and the ellipse and the auxiliary circle at the points P and Q respectively.

- (i) Find the equations of the tangent to the ellipse at P and to the auxiliary circle at Q. (4)
 - (ii) The tangents at P and Q intersect at point R. Show that R lies on the x axis. (2)
 - (iii) Prove that $ON \cdot OR$ is independent of the positions of P and Q. (1)
- (b) On Tuesday morning at 5 am, a truck crashes into a harbour. The rescue team and their equipment can only work effectively when the depth of water is no more than 7 m. At low tide, the depth of the water is 5 metres and at high tide, the depth is 10 metres. Low tide occurs at 4 am and high tide at 10.15 am. Assume the movement of the tide is simple harmonic motion.
- (i) Find the period and amplitude of the motion. (2)
 - (ii) If the deadline for the rescue operation is 6 pm on Wednesday evening, find the periods of time between 5 am and 6 pm during which the rescue team can work effectively. (4)
- (c) Given a real polynomial $Q(x)$ show that if α is a root of $Q(x)-x=0$, then α is also a root of $Q(Q(x))-x=0$. (2)

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Question 7 (Answer in a separate booklet)

(15)

(a) For the curve $x^2y^2 - x^2 + y^2 = 0$

(i) state any x and y -intercepts. (1)

(ii) demonstrate why $|y| < 1$. (1)

(iii) demonstrate why $|y| \leq |x|$. (1)

(iv) use implicit differentiation to show $\frac{dy}{dx} = \frac{x(1-y^2)}{y(1+x^2)}$. (2)

(v) state the co-ordinates of any critical points (where the derivative is undefined). (1)

(vi) hence, explain why the curve has no stationary points. (1)

(vii) state the horizontal asymptote(s). (1)

(viii) sketch the curve. (1)

(b) By expanding $(\cos \theta + i \sin \theta)^3$ it can be shown that

$$\cot 3\theta = \frac{t^3 - 3t}{3t^2 - 1} \text{ where } t = \cot \theta$$

(i) solve $\cot 3\theta = -1$ for $0 \leq \theta \leq 2\pi$ (2)

(ii) Hence show that $\cot \frac{\pi}{12} \cdot \cot \frac{5\pi}{12} \cdot \cot \frac{9\pi}{12} = -1$ (2)

(iii) Write down a cubic equation with roots $\tan \frac{\pi}{120}$, $\tan \frac{5\pi}{12}$ and $\tan \frac{9\pi}{12}$. (2)

(Express your answer as a polynomial equation with positive integer coefficients).

Question 8 (Answer in a separate booklet)

(15)

(a) The series $\frac{1}{2} + \frac{8}{4} + \frac{27}{8} + \dots = \sum_{n=1}^{\infty} \frac{n^3}{2^n}$ is not geometric and, as such, it is not a routine matter to decide whether or not it converges to a finite sum. Let $y_n = \frac{n^3}{2^n}$

(i) Show that $\frac{y_n}{y_{n-1}} = \frac{1}{2} \times \left(\frac{n}{n-1}\right)^3$ and hence show that this ratio is greater than 1 when $2 \leq n \leq 4$ but less than 1 when $n \geq 5$. (3)

(ii) Show that $\frac{y_n}{y_{n-1}} \leq 0.98$ for $n \geq 5$. (2)

(iii) Given that $y_4 = 4$, deduce that $y_n \leq 4 \times (0.98)^{n-4}$ for $n \geq 4$ and write down the value of $\lim_{n \rightarrow \infty} y_n$. (2)

(b) Show that the derivative of the function $y = x^x$ for $x > 0$ is $(\log_e x + 1) x^x$. (2)

(c) Find all solutions ($0 \leq \theta < \pi$) in radians of the equation $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} = \frac{3}{4}$. (3)

(d) A triangle ABC is right-angled at A and it has sides of lengths a, b and c units (the side opposite angle A is a etc). A circle of radius r units is drawn so that the sides of the triangle are tangents to the inscribed circle.

Prove that $r = \frac{1}{2}(c + b - a)$. (3)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1:

(a)	Let $u=2x$ $du = 2dx$ $\int \frac{1}{\sqrt{4x^2 - 25}} dx$ $= \int \frac{1}{\sqrt{u^2 - 5^2}} \frac{du}{2}$ $= \frac{1}{2} \ln(u + \sqrt{u^2 - 5^2}) + c$ $= \frac{1}{2} \ln(2x + \sqrt{4x^2 - 25}) + c$ Alternatively $\int \frac{1}{\sqrt{4x^2 - 25}} dx$ $= \int \frac{1}{\sqrt{4\left(x^2 - \frac{25}{4}\right)}} dx$ $= \frac{1}{2} \int \frac{1}{\sqrt{x^2 - \left(\frac{5}{2}\right)^2}} dx$ $= \frac{1}{2} \ln\left(x + \sqrt{x^2 - \frac{25}{4}}\right) + c_1$ NB These solutions only differ by a constant of integration as $\frac{1}{2} \ln\left(x + \sqrt{x^2 - \frac{25}{4}}\right) + c_1$ $= \frac{1}{2} \ln\left(x + \sqrt{\frac{4x^2 - 25}{4}}\right) + c_1$ $= \frac{1}{2} \ln\left(\frac{2x + \sqrt{4x^2 - 25}}{2}\right) + c_1$ $= \frac{1}{2} \ln(2x + \sqrt{4x^2 - 25}) - \frac{1}{2} \ln 2 + c_1$ $\therefore c = c_1 - \frac{1}{2} \ln 2$	2 marks – correct solution
(b)	$\int \tan^4 x \sec^2 x dx$ $= \int \tan^4 x d(\tan x)$ $= \frac{\tan^5 x}{5} + c$	2 marks – correct solution 1 mark – applying the standard integral.

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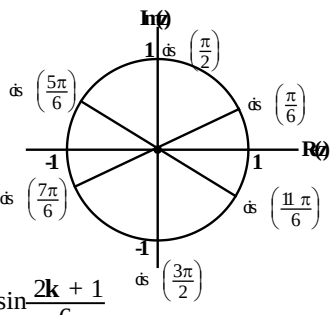
(c)	$\int_0^{\frac{\pi}{2}} x \sin x \, dx$ $= \left[-x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \, dx$ $= (0 - -0) + \left[\sin x \right]_0^{\frac{\pi}{2}}$ $= 0 + (1 - 0)$ $= 1$	3 marks – correct solution. 2 marks – correctly integrating $\sin x$. 1 mark – correct substitution into integration by parts process.
(d)	$u = x - 2$ $u + 1 = x - 1$ $1 - u = 3 - x$ when $x = \frac{3}{2}$, $u = -\frac{1}{2}$ when $x = \frac{5}{2}$, $u = \frac{1}{2}$ $\int_{\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{(1+u)(1-u)}} \, du$ $= \int_{\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-u^2}} \, du$ $= \left[\sin^{-1} u \right] - \frac{1}{2}$ $= \frac{\pi}{6} - -\frac{\pi}{6}$ $= \frac{\pi}{3}$	3 marks – correct solution. 2 marks – correctly integrating. 1 mark – correct change of variable.

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(e) i	$\frac{3x+2}{(x+3)(x+2)} \equiv \frac{A}{x+3} + \frac{B}{x+2}$ $3x+2 = A(x+2) + B(x+3)$ When $x = -2$, $-4 = B \therefore B = -4$ When $x = -3$, $-7 = -A \therefore A = 7$ $\frac{3x+2}{(x+3)(x+2)} \equiv \frac{7}{x+3} - \frac{4}{x+2}$	2 marks – correct solution 1 marks – finding A or B.
(e) ii	$\int_0^2 \frac{3x+2}{x^2+5x+6} dx$ $= \int_0^2 \frac{7}{x+3} - \frac{4}{x+2} dx$ $= \left[7\ln(x+3) - 4\ln(x+2) \right]_0^2$ $= 7\ln 5 - 4\ln 4 - 7\ln 3 + 4\ln 2$ $= 7\ln\left(\frac{5}{3}\right) - 4\ln 2$	3 marks – correct solution. 2 marks – correct substitution. 1 mark – correct integrand.

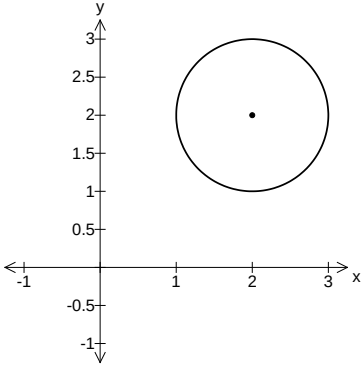
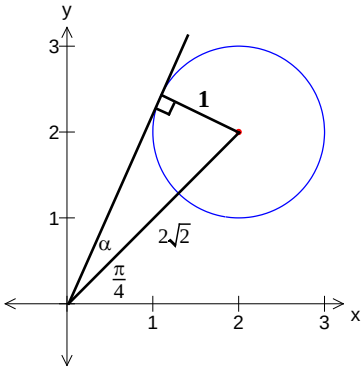
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Question 2

(a)	$z = \sqrt{2} - \sqrt{2}i = 2 \operatorname{cis} \left(-\frac{\pi}{4} \right)$ $\therefore z^{10} = 2^{10} \operatorname{cis} \left(\frac{-10\pi}{4} \right) = -2^{10}i = -1024i$	2 marks – correct answer 1 mark – correct mod and arg for z
(b)(i)	$(O - A)(-i) = P - A$ $P = A + Ai$ $= z(1 + i)$	1 mark – correct vector for P
(ii)	$\frac{(O - A)i}{2} = M - A$ $\therefore M = z \left(\frac{i}{2} + 1 \right)$	1 mark – correct vector for M
(iii)	OC must be // AB $k(A - O)I = C - O \quad (k \neq 1)$ $C = kzi - eg \ 2zi$	1 mark – correct vector
(c(i))	$i^6 = i^4 * i^2$ $= (1)(-1) = -1$	1 mark – correct value
(ii)	$z^6 = -1 + 0 \times i$ $= (\cos \theta + i \sin \theta)^6$ $= \cos 6\theta + i \sin 6\theta$ $\therefore 6\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi, \dots$ $\theta = \frac{\pi}{6}, \frac{3\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}$ $\therefore \text{sixth roots are } \cos \frac{2k+1}{6}\pi + i \sin \frac{2k+1}{6}\pi$ for k = 0, 1, 2, 3, 4, 5 	2 marks – correct solutions and correct diagram appropriately labelled 1 mark - correct solutions but incompletely labelled
(c)(iii)	$z^6 + 1 = 0$ $(z^2)^3 + (1)^3 = 0$ $(z^2 + 1)(z^4 - z^2 + 1) = 0$ But $z^2 + 1 = 0$ has roots $z = \pm i$ $\therefore z^4 - z^2 + 1 = 0 \text{ has roots}$ $z = \operatorname{cis} \frac{\pi}{6}, \operatorname{cis} \frac{5\pi}{6}, \operatorname{cis} \frac{7\pi}{6} \text{ and } \operatorname{cis} \frac{11\pi}{6}$	2 marks – correct roots 1 mark – recognition of the basic factorisation

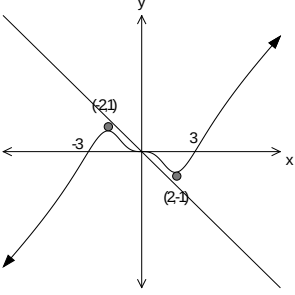
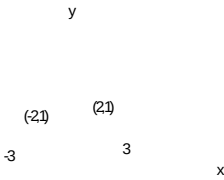
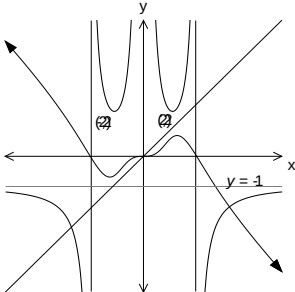
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Question 2 (continued)

(d) (i)		1 mark – correct diagram showing the locus (only)
(ii)	Max arg when the tangent is on left side of the circle) - NOT through point (1, 2). $\therefore \sin \alpha = \frac{1}{2\sqrt{2}}$ $\therefore \alpha = 20^{\circ}42'$ argument of vector to centre is $\frac{\pi}{4}$ $\therefore \text{max arg} = \frac{\pi}{4} + \sin^{-1}\left(\frac{1}{2\sqrt{2}}\right) = 65^{\circ}42'$ 	2 marks – correct argument 1 mark – correct argument for vector to centre
(iii)	Max modulus is when the vector extends through the centre to the other side of the circle. So modulus vector is $2\sqrt{2}$ and radius is 1. Hence max modulus is $2\sqrt{2} + 1$	2 marks – correct modulus 1 mark – indication of where the modulus would be

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Question 3:

(a)i		1 mark –correct graph.
(a)ii		1 mark –correct graph.
(a)iii		3 mark –correct graph. 1 for both branches above the x-axis 1 for both branches below x-axis 1 for horizontal asymptote at $y = -1$ and for turning points at $(\pm 2, 2)$

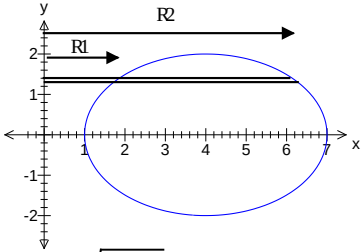
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(b)	As $(1+i)$ is a zero, $(1-i)$ is also a zero. $\therefore (x - (1 + i))(x - (1 - i))$ is a factor $= (x^2 - 2x + 2)$ so other factors maybe found by factor theorem or polynomial division $P(1) = 0 \therefore x - 1$ is a factor $P(-2) = 0 \therefore x + 2$ is a factor $\therefore P(x) = (x - 1)(x + 2)(x - 1 - i)(x - 1 + i)$	2 mark – correct factorisation 1 mark- correct application of “complex roots of a polynomial, with real coefficients, occur in conjugate pairs”
(c) i	$x^3 + x^2 - 2x - 3 = 0$ has roots $x = \alpha, \beta$ and γ So the equation with roots $X = \alpha^2, \beta^2$ and γ^2 will have $\sqrt{X} = \alpha, \beta$ and γ so $x = \sqrt{X}$ will satisfy the original equation. $(\sqrt{X})^3 + (\sqrt{X})^2 - 2(\sqrt{X}) - 3 = 0$ $X^{\frac{3}{2}} + X - 2X^{\frac{1}{2}} - 3 = 0$ $X^{\frac{3}{2}} - 2X^{\frac{1}{2}} = 3 - X$ $X^3 - 4X^2 + 4X = 9 - 6X + X^2$ $X^3 - 5X^2 + 10X - 9 = 0$	1 mark correct equation.
(c) ii	$\alpha \beta \gamma = 3$ $\therefore \alpha^3 \beta \gamma, \alpha \beta^3 \gamma, \alpha \beta \gamma^3$ $= \alpha \beta \gamma \alpha^2, \alpha \beta \gamma \beta^2, \alpha \beta \gamma \gamma^2$ $= 3\alpha^2, 3\beta^2, 3\gamma^2$ which are 3 times the roots of $x^3 - 5x^2 + 10x - 9 = 0$ So substitute $\frac{x}{3}$ into $x^3 - 5x^2 + 10x - 9 = 0$ Alternatively, substitute $\sqrt{\frac{x}{3}}$ into $x^3 + x^2 - 2x - 3 = 0$ $\left(\frac{x}{3}\right)^3 - 5\left(\frac{x}{3}\right)^2 + 10\left(\frac{x}{3}\right) - 9 = 0$ $\frac{x^3}{27} - \frac{5x^2}{9} + \frac{10x}{3} - 9 = 0$ $x^3 - 15x^2 + 90x - 243 = 0$	2 marks correct polynomial 1 mark for obtaining $3\alpha^2, 3\beta^2, 3\gamma^2$

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(d) i	$b^2 = a^2(e^2 - 1)$ $25 = 16(e^2 - 1)$ $e^2 = 1 + \frac{25}{16}$ $e = \pm \frac{\sqrt{41}}{4}$ but, as for a hyperbola, $e > 0$ then $e = \frac{\sqrt{41}}{4}$	1 mark
(d) ii	$(\pm ae, 0) = (\pm \sqrt{41}, 0)$	1 mark
(d) iii	$x = \pm \frac{a}{e} = \pm \frac{16}{\sqrt{41}}$	1 mark
(d) iv	$y = \pm \frac{b}{a}x$ $y = \pm \frac{5}{4}x$	1 mark
(d) v	$ PS - PS' = 2a$ $ 2 - PS' = 8$ $PS' = 10$ or -6 , but PS' is a distance $PS' = 10$	1 mark

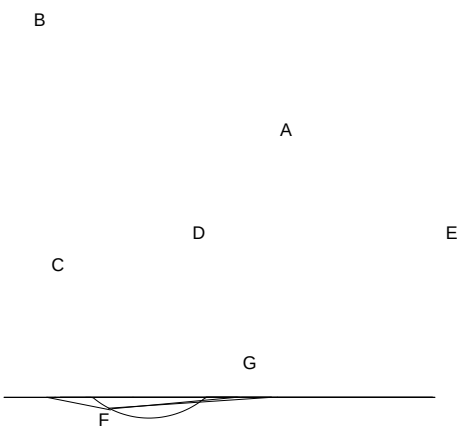
Question 4

(a)	Area of shell = $\pi\{(x + \delta x)^2 - x^2\}$ $\therefore \text{Vol} = \pi 2x \, dx$ after letting $\delta x^2 \rightarrow 0$ or $2\pi xy \, \delta x = 2\pi x \ln x \, \delta x$ $\therefore \text{Vol} = \lim_{\delta x \rightarrow 0} \sum_1^e 2\pi x \ln x \, \delta x$ $= 2\pi \int_1^e x \ln x \, dx$ $= 2\pi \left[\ln x \times \frac{x^2}{2} \right]_1^e - 2\pi \int_1^e \frac{x}{2} \, dx$ $= 2\pi \left[\frac{e^2}{2} - 0 \right] - 2\pi \left[\frac{e^2}{4} - \frac{1}{4} \right]$ $= \frac{\pi}{2} [e^2 + 1]$	4 marks – correct volume 3 marks – application of integration by parts but subsequent error 2 marks – correct statement for V 1 mark – correct development for δV. NOTE: $(x-1)$ is not necessary because the integral goes from 1 to e.
(b) (i)	$\delta V = \pi (R_2^2 - R_1^2) \delta y$ $x = 4 \pm 3\sqrt{1 - \frac{y^2}{4}}$ $\therefore R_2 + R_1 = 8$ and $R_2 - R_1 = 6\sqrt{1 - \frac{y^2}{4}}$ $\delta V = \pi 8 \times \frac{6}{2} \times \sqrt{4 - y^2} \delta y$ $= 8\pi \sqrt{36 - 9y^2}$ 	2 marks – correct demonstration 1 mark – correct approach with sufficient progress
(ii)	$V = \lim_{\delta y \rightarrow 0} \sum_{-2}^2 24\pi \sqrt{4 - y^2} \delta y$ $= 24\pi \int_{-2}^2 \sqrt{4 - y^2} \, dy$ $= 24\pi \times \frac{\pi \times 2^2}{2} = 48\pi^2$	2 marks – correct volume calculated 1 mark – correct approach using any appropriate method (eg trig substitution). Always make your life easier by looking for the "circle approach" when you can.

Question 4 (continued)

(c)	Sum of roots = $\frac{-(2-i)}{3+4i}$ $= \frac{(i-2)}{3+4i} \times \frac{3-4i}{3-4i}$ $= \frac{11i-2}{25}$ $\therefore \text{Modulus} = \sqrt{\frac{11^2 + 2^2}{25^2}} = \frac{1}{\sqrt{5}} = 0.447$ Argument = $\tan^{-1}\left(-\frac{11}{25}\right) = \tan^{-1}\left(-\frac{11}{2}\right) = 100^\circ 18'$	2 marks – correct answers 1 mark – sum of roots expressed correctly
(d)(i)	Let the points be P $\left(cp, \frac{c}{p}\right)$ and Q $\left(cq, \frac{c}{q}\right)$ $\therefore \frac{\frac{y-c}{p}}{\frac{c}{q} - \frac{c}{p}} = \frac{x-cp}{cq - cp}$ $\frac{py-c}{\frac{c(p-q)}{pq}} = \frac{x-cp}{c}$ $\frac{(py-c)q}{c} = -\frac{x-cp}{c}$ $\therefore x + pqy = c(p+q)$	2 marks – correct proof 1 mark – correct approach with progress but minor error
(ii)	$PQ^2 = (cp - cq)^2 + \left(\frac{c}{p} - \frac{c}{q}\right)^2 = k^4$ $= c^2(p-q)^2 + \frac{c^2}{p^2q^2}(q-p)^2$ $\therefore c^2(p-q)^2 \left(1 + \frac{1}{p^2q^2}\right) = k^4$ $\therefore c^2[(p+q)^2 - 4pq](p^2q^2 + 1) = k^4 p^2 q^2$ R is $X = \frac{cp+cq}{2} = \frac{c(p+q)}{2}$ $Y = \frac{cq+cp}{2pq}$ $\therefore$ eliminating $(p+q)$ $X = pqY$ Substituting the $X = \text{eqn}$ for $(p+q)$ and the last for pq into the previously proved equation gives: $c^2 \left[\frac{4x^2}{c^2} - \frac{4x}{y} \right] \left[\frac{x^2}{y^2} + 1 \right] = k^4 \frac{x^2}{y^2}$ $[4x^2y - 4xc^2][x^2 + y^2] = k^4 x^2 y$ $4(xy - c^2)(x^2 + y^2) = k^4 xy$	3 marks – final equation for locus correct 2 marks – values for x and y calculated correctly and some appropriate progress made from there 1 mark – first equation proved correctly

Question 5:

(a) i	 Construct chords DG, GF and GE as shown $\angle DGF = \angle BCD$ (Exterior angle of cyclic quadrilateral equals interior opposite angle) $\angle DGE = \angle BAD$ (Exterior angle of cyclic quadrilateral equals interior opposite angle) $\angle BCD + \angle BAD = 180$ (opposite angles of cyclic quadrilateral are supplementary) $\therefore \angle DGF + \angle DGE = 180$ Hence F, G and E are collinear, as angles on a straight line are supplementary.	4 marks – 1 mark for each correct step with a reason.
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(b) i	$I_n = \int_0^{\frac{\pi}{2}} \sin^{n-1} x \sin x \, dx$ $u = \sin^{n-1} x \quad dv = \sin x$ $du = (n-1) \sin^{n-2} x \cos x \, dx \quad v = -\cos x$ $I_n = \left[\sin^{n-1} x \times -\cos x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x \cos x \times -\cos x \, dx$ $= [0 - 0] + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$ $= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$	2 marks – correct solution 1 mark – correct choice of u and dv .
(b) ii	$I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$ $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) \, dx$ $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x - \sin^n x \, dx$ $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ $I_n = (n-1) I_{n-2} - (n-1) I_n$ $I_n + (n-1) I_n = (n-1) I_{n-2}$ $n I_n = (n-1) I_{n-2}$ $I_n = \frac{n-1}{n} I_{n-2}$	2 marks – correct proof. 1 marks – for obtaining $(n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x - \sin^n x \, dx$
(b) iii	$I_4 = \frac{3}{4} I_2$ $= \frac{3}{4} \left(\frac{1}{2} I_0 \right)$ $= \frac{3}{8} \int_0^{\frac{\pi}{2}} \sin^0 x \, dx$ $= \frac{3}{8} \left[x \right]_0^{\frac{\pi}{2}}$ $= \frac{3\pi}{16}$	2 marks – correct solution. 1 mark – correct primitive
(c) i	Let $P(x) = (x - \alpha)^n Q(x)$ $P'(x) = n(x - \alpha)^{n-1} Q(x) + (x - \alpha)^n Q'(x)$ $P'(x) = (x - \alpha)^{n-1} [nQ(x) + (x - \alpha)Q'(x)]$ $\therefore P'(\alpha) = 0 \text{ and } \alpha \text{ is a root of } P'(x)$	2 marks – correct proof. 1 mark for defining $P(x)$ as Let $P(x) = (x - \alpha)^n Q(x)$

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(c) ii	$P'(x) = 54x^2 + 6x - 28$ $P'(x) = 2(27x^2 + 3x - 14)$ $P'(x) = 2(9x + 7)(3x - 2)$ $x = -\frac{7}{9} \text{ or } \frac{2}{3}$ $P\left(\frac{2}{3}\right) = 18\left(\frac{2}{3}\right)^3 + 3\left(\frac{2}{3}\right)^2 - 28\left(\frac{2}{3}\right) + 12 = 0$ $\therefore P\left(\frac{2}{3}\right) = P'\left(\frac{2}{3}\right) = 0$ and $x=2/3$ is the double root. $\therefore (3x - 2)^2(ax - \beta) = 18x^3 + 3x^2 - 28x + 12$ $(9x^2 - 12x + 4)(ax - \beta) = 18x^3 + 3x^2 - 28x + 12$ so by inspection of the coefficients $(3x - 2)^2(2x + 3) = 18x^3 + 3x^2 - 28x + 12$ the roots are $2/3$ and $-3/2$	3 marks – correct solution. 2 mark – $P'\left(\frac{2}{3}\right) = P\left(\frac{2}{3}\right) = 0$ 1 mark- $x = -\frac{7}{9} \text{ or } \frac{2}{3}$
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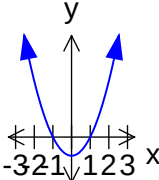
Question 6

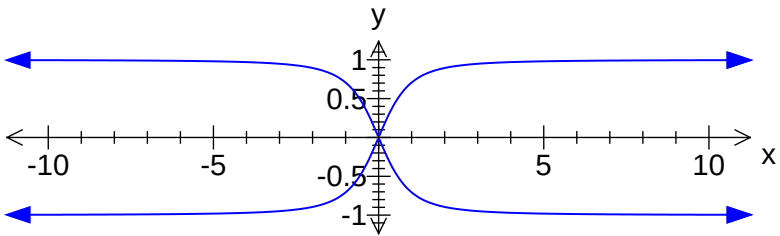
(a)(i)	At P $x = 4 \cos \theta$ $y = 3 \sin \theta$ $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$ $= \frac{-3 \cos \theta}{4 \sin \theta}$ $\therefore y - 3 \sin \theta = \frac{-3 \cos \theta}{4 \sin \theta} (x - 4 \cos \theta)$ $\frac{x \cos \theta}{4} + \frac{y \sin \theta}{3} = 1$ At Q $x = 4 \cos \theta$ $y = 4 \sin \theta$ $\therefore \frac{dy}{dx} = \frac{-\cos \theta}{\sin \theta}$ $y - 4 \sin \theta = \frac{-\cos \theta}{\sin \theta} (x - 4 \cos \theta)$ $\frac{x \cos \theta}{4} + \frac{y \sin \theta}{4} = 1$	For each equation: 2 marks – equation correct 1 mark – gradient correct
(ii)	Solving simultaneously gives $x \cos \theta = 4$ $\therefore$ substituting back in $16 + 4y \sin \theta = 16$ $\therefore y = 0$ (so the x - axis)	2 marks – correct demonstration 1 mark – correct approach to solving simultaneous equations
(iii)	ON = $4 \cos \theta$ OR = $\frac{4}{\cos \theta}$ $\therefore$ ON $\times$ OR = 16 which is independent of θ and so of the positions of P and Q	1 mark – correct demonstration
(b) (i)	Amplitude = 2.5 m Period = 12.5 hours	2 marks – both amplitude and period correct 1 mark – either amplitude and period correct

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(ii)	$n = \frac{4\pi}{25}$ $\therefore x = 2.5 \cos\left(\frac{4\pi t}{25} + \varepsilon\right)$ $t = 0, x = -2.5 \text{ (so at low tide)} \therefore \varepsilon = \pi$ $\therefore x = 2.5 \cos\left(\frac{4\pi t}{25} + \pi\right) = -2.5 \cos\left(\frac{4\pi t}{25}\right)$ $x = -0.5 t = \frac{25}{4\pi} \cos^{-1}(0.2)$ $\therefore t = 2 \text{ hrs } 43 \text{ mins after 4 am i.e at 6 hrs } 43 \text{ mins}$ and 2hr 43 mins before 4:30 pm - so 1:47 pm so crew can work from 5 am to 6:43 am and from 1:47 pm to 6:00 pm	4 marks – correct time intervals determined 3 marks – correct expression for time 2 marks – correct value for initial phase determined 1 mark – appropriate format for a cos equation stated
(c)	If α is a root $Q(\alpha) - \alpha = 0$ $\therefore Q(\alpha) = \alpha$ $\therefore$ when α is substituted $Q(Q(x)) - x = Q(Q(\alpha)) - \alpha$ $= Q(\alpha) - \alpha = 0$ $\therefore \alpha$ is a root	2 marks – correct demonstration 1 mark – some appropriate substitution made.

Question 7:

(a) i	When $x = 0$ $0 - x^2 - 0 = 0$ $\therefore x = 0$ So x-y intercept is $(0, 0)$	1mark
ii	$x^2 y^2 - x^2 + y^2 = 0$ rearranges to $y^2 = \frac{x^2}{1+x^2}$ and since $x^2 \leq 1+x^2$ $y^2 < 1$ $y^2 - 1 < 0$ $(y-1)(y+1) < 0$ $-1 < y < 1$ $y < 1$ 	1mark
iii	The equation may also be rearranged to $\frac{y^2}{x^2} = 1 - y^2$ As $y^2 \geq 0$ $-y^2 \leq 0$ then $1 - y^2 \leq 1$ $\therefore \frac{y^2}{x^2} \leq 1$ $y^2 \leq x^2$ $\sqrt{y^2} \leq \sqrt{x^2}$ $y \leq x$	1mark
iv	$2xy^2 + x^2 2y \frac{dy}{dx} - 2x + 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx}(2x^2 y + 2y) = 2x - 2xy^2$ $\frac{dy}{dx} = \frac{2x - 2xy^2}{2x^2 y + 2y}$ $\frac{dy}{dx} = \frac{2x(1 - y^2)}{2y(1 + x^2)}$ $\frac{dy}{dx} = \frac{x(1 - y^2)}{y(1 + x^2)}$	2 marks for correct demonstration 1 mark for implicit differentiation
v	Critical points occur when $\frac{dy}{dx}$ is undefined at $y=0$. Therefore the critical point is $(0,0)$.	1mark

vi	Possible stationary points occur when $\frac{dy}{dx} = 0$, $0 = \frac{x(1-y^2)}{y(1+x^2)}$ $x = 0$ or $y = \pm 1$ But these values do not result in stationary points as they are either at the critical point or outside the range determined above. So there can be no stationary points.	1 mark
vii	$y = -1$ Use limit as x approaches plus or minus infinity.	1 mark
viii	Note the relation can be rewritten in form $y^2 = \frac{x^2}{1+x^2}$ and all curves of form $y^2 = f(x)$ have symmetry about x-axis because $y = \pm \frac{x}{\sqrt{1+x^2}}$. Also it is odd. 	1 mark
b i	$\cot 3\theta = -1$ $\tan 3\theta = -1$ $3\theta = -\frac{\pi}{4} + k\pi$ where k is an integer $3\theta = \frac{-\pi + 4k\pi}{4}$ $\theta = \frac{-\pi(1-4k)}{12}$ so for $0 \leq \theta < 2\pi$ when $k=1,2,3,4,5,6$ $\theta = \frac{\pi}{4}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$	2 marks for correct angles. 1 mark for some of the correct angles.

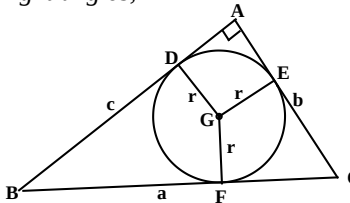
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	If $\cot 3\theta = -1$ and given $\cot 3\theta = \frac{t^3 - 3t}{3t^2 - 1}$ then $-1 = \frac{t^3 - 3t}{3t^2 - 1}$ $1 - 3t^2 = t^3 - 3t$ $0 = t^3 + 3t^2 - 3t - 1$ 3 unique solutions of this polynomial are $t = \cot \frac{\pi}{4}, \cot \frac{7\pi}{12} \text{ and } \cot \frac{11\pi}{12}$ By the product of the roots of $0 = t^3 + 3t^2 - 3t - 1$ $\cot\left(\frac{\pi}{4}\right) \cot\left(\frac{7\pi}{12}\right) \cot\left(\frac{11\pi}{12}\right) = 1$ and since $\cot(-\theta) = -\cot\theta$ then $-\cot\left(\frac{3\pi}{4}\right) \times -\cot\left(\frac{5\pi}{12}\right) \times -\cot\left(\frac{\pi}{12}\right) = 1$ $-\cot\left(\frac{9\pi}{12}\right) \times \cot\left(\frac{5\pi}{12}\right) \times \cot\left(\frac{\pi}{12}\right) = 1$ $\cot\left(\frac{\pi}{12}\right) \times \cot\left(\frac{5\pi}{12}\right) \times \cot\left(\frac{9\pi}{12}\right) = -1$	2 marks for correct demonstration. 1 mark for a <u>correct</u> statement regarding the product of the roots.
	$0 = \left(\frac{1}{t}\right)^3 + 3\left(\frac{1}{t}\right)^2 - 3\left(\frac{1}{t}\right) - 1$ $0 = 1 + 3t - 3t^2 - t^3$	1 mark for correct demonstration

Question 8:

(a) (i)	$f(n) = \frac{y_n}{y_{n-1}} = \frac{n^3}{2^n} \times \frac{2^{n-1}}{(n-1)^3}$ $= \frac{1}{2} \left(\frac{n}{n-1} \right)^3$ $f'(n) = \frac{3}{2} \left(\frac{n}{n-1} \right)^2 \times \frac{(n-1) \times 1 - n \times 1}{(n-1)^2}$ $= \frac{-3n^2}{2(n-1)^4}$ For $n \geq 5$ $f'(n) < 0$ so it is a decreasing function $f(2) = 4$ $f(3) = \frac{27}{16}$ and $f(4) = \frac{32}{27}$ but $f(5) = \frac{125}{128} < 1$ $\therefore$ ratio > 1 for $2 \leq n \leq 4$ but < 1 for $n \geq 5$	3 marks – correct demonstration or interpretation of pattern 2 marks – progress with the derivative interpretation or examination of values 1 mark – correct simplification of ratio and then an attempt to apply it NOTE: you need a strategy to show that the ratio does not start to increase.
(ii)	We know that $f(5) = \frac{125}{128} = 0.977 < 0.98$ and the function is decreasing $\therefore$ always less than 0.98	2 marks – correct argument 1 mark – some justification as to why it should be less than 0.98
(iii)	The ratio of successive terms in this series is less than 0.98 (from part (ii)). Hence the series will decrease more rapidly than a geometric series which has a common ratio of 0.98. So $y_n \leq 4 \times (0.98)^{n-4}$ for $n \geq 4$ as $y_4 = 4$. But for the GS, the nth term approaches 0 because $(0.98)^{n-4}$ approaches 0. So the limit of y_n is zero.	2 marks – correct argument 1 mark – correct approach but logic error
(b)	$y = x^x$ $\therefore \ln y = x \ln x$ $y = e^{x \ln x}$ $\frac{d}{dx}(x \ln x) = \ln x + 1$ $\therefore \frac{dy}{dx} = e^{x \ln x} \times (\ln x + 1)$ $= (\ln x + 1) x^x$	2 marks – correct demonstration 1 mark – transformation to base e

Question 8 (continued)

(c)	$\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} = \frac{3}{4}$ $\frac{(\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)}{\sin \theta + \cos \theta} = \frac{3}{4}$ $\therefore 1 - \sin \theta \cos \theta = \frac{3}{4}(\sin \theta - \cos \theta)$ $\therefore \sin 2\theta = \frac{1}{2}$ $2\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ $\theta = \frac{\pi}{12}, \frac{5\pi}{12}$	3 marks – two correct solutions 2 marks – relevant progress to a statement for 2θ. 1 mark – numerator factored correctly
(d)	As radii meet the tangents at right angles, $\angle ADG = \angle AEG = 90^\circ$ As $\angle DAE = 90^\circ$ (given) and $AD = AE$ (tangents from an external point are equal) AEGD is a square. So $AD = AE = r$ So $BD = c - r = BF$ $EC = b - r = CF$ So $BC = a = b - r + c - r$ $\therefore r = \frac{1}{2}(c + b - a)$	 3 marks – correct proof 2 marks – appropriate work towards solution 1 mark – identification that AD and $AE = r$